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Theopractical Sheet 4
FAT & NTFS

Question Block 1 (6 points)
Analyze the given image file. While Autopsy might help you for some subquestions, you will also need hexdump/xxd/hexedit/fsstat for some other questions.

Question 1.1 (0.5 point)
What is the drive name of the given partition? Give it as a string.

Ans:

```
┌──(kali㉿kali)-[~/Downloads/Theopractical Sheet #4 (2)/Theopractical Sheet #4]
└─$ fsstat image.dd
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: FAT16

OEM Name: mkfs.fat
Volume ID: 0x7093d7b8
Volume Label (Boot Sector): FLAG{S12H9}
Volume Label (Root Directory): FLAG{S12H9}
File System Type Label: FAT32

Sectors before file system: 2048

File System Layout (in sectors)
Total Range: 0 - 18431
* Reserved: 0 - 3
** Boot Sector: 0
* FAT 0: 4 - 23
* FAT 1: 24 - 43
* Data Area: 44 - 18431
** Root Directory: 44 - 75
** Cluster Area: 76 - 18431

METADATA INFORMATION
--------------------------------------------
Range: 2 - 294214
Root Directory: 2

CONTENT INFORMATION
--------------------------------------------
Sector Size: 512
Cluster Size: 2048
Total Cluster Range: 2 - 4590

FAT CONTENTS (in sectors)
--------------------------------------------
80-91 (12) -> 96
92-95 (4) -> EOF
96-171 (76) -> EOF
172-215 (44) -> EOF
216-1831 (1616) -> EOF
1832-1835 (4) -> EOF
1836-1839 (4) -> EOF
1840-1843 (4) -> EOF
```

The Volume Label is the drive name. The output from fsstat showed me:
Boot Sector Volume Label: FLAG{S12H9}
Root Directory Volume Label: FLAG{S12H9}
From this I got The drive name is FLAG{S12H9} since both match. The volume label in FAT is usually kept in the root directory or boot sector.

Question 1.2 (0.5 points)
Is this a FAT12, FAT16 or FAT32 partition? Give it as a string.

Ans: From fsstat output we can clearly see that it explicitly states that File System Type:FAT 16.

Question 1.3 (0.5 points)
At which sector starts the root directory entry? Give the result as a decimal number

Ans:The fsstat output clearly states as shown in the screenshot provided above that the root directory :44-75
Hence 44

Question 1.4 (0.5 point)
Give the flag from the text file with the two last characters replaced as instructed in the text file

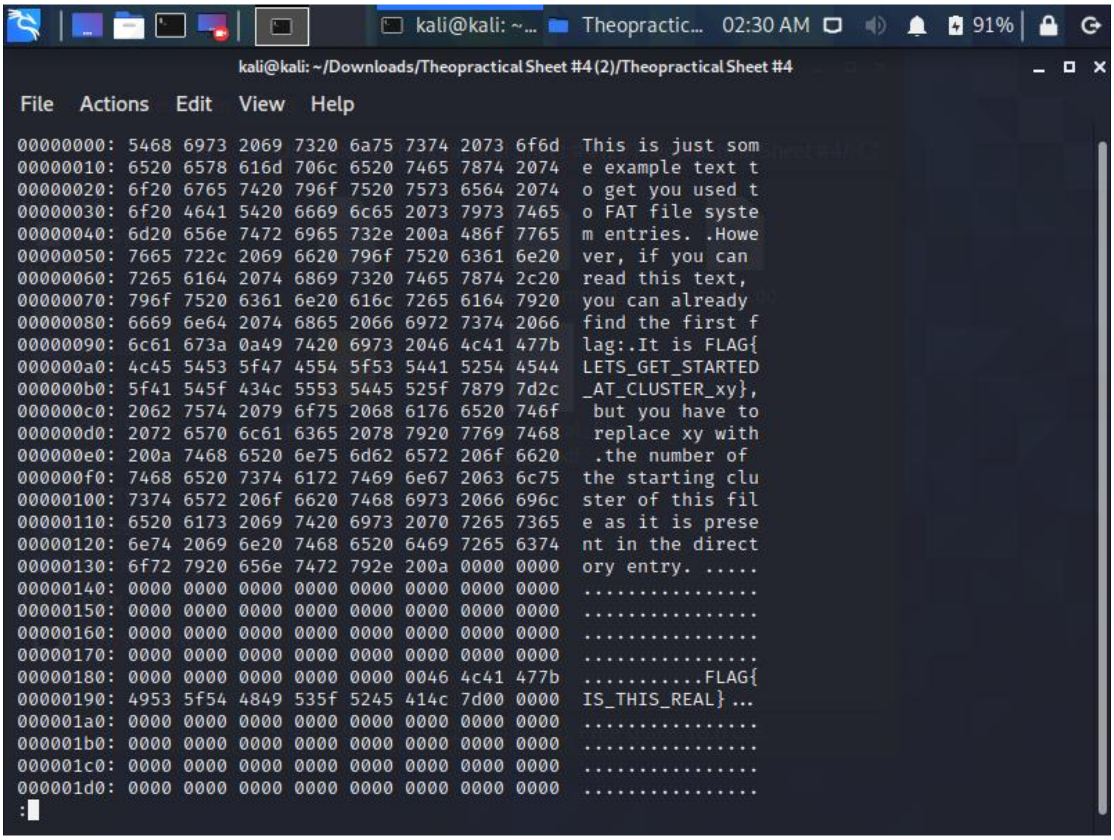
Ans From fsstat I got the following:
Sectors per Cluster: 4 (since Cluster Size = 2048 bytes and Sector Size = 512 bytes, 2048 / 512 = 4).
First Data Sector: 76(start of Data Area).
Starting Sector for file1.txt: 92 (from metadata in autopsy).

Starting Cluster = (Starting Sector - First Data Sector) / Sectors per Cluster + 2
(92 - 76) / 4 + 2 = 16 / 4 + 2 = 4 + 2 = 6

The starting Cluster is 14.
FLAG{LETS_GET_STARTED_AT_CLUSTER_06}

Question 1.5 (1 point)
Find the flag in the slack space of the text file (0.5 points). Is it part of the RAM Slack or the Drive Slack? (0.5 points)

Ans:




```
00000000: 5468 6973 2069 7320 6a75 7374 2073 6f6d  This is just som
00000010: 6520 6578 616d 706c 6520 7465 7874 2074  e example text t
00000020: 6f20 6765 7420 796f 7520 7573 6564 2074  o get you used t
00000030: 6f20 4641 5420 6669 6c65 2073 7973 7465  o FAT file syste
00000040: 6d20 656e 7472 6965 732e 200a 486f 7765  m entries. .Howe
00000050: 7665 722c 2069 6620 796f 7520 6361 6e20  ver, if you can
00000060: 7265 6164 2074 6869 7320 7465 7874 2c20  read this text,
00000070: 796f 7520 6361 6e20 616c 7265 6164 7920  you can already
00000080: 6669 6e64 2074 6865 2066 6972 7374 2066  find the first f
00000090: 6c61 673a 0a49 7420 6973 2046 4c41 477b  lag:.It is FLAG{
000000a0: 4c45 5453 5f47 4554 5f53 5441 5254 4544  LETS_GET_STARTED
000000b0: 5f41 545f 434c 5553 5445 525f 7879 7d2c  _AT_CLUSTER_xy},
000000c0: 2062 7574 2079 6f75 2068 6176 6520 746f   but you have to
000000d0: 2072 6570 6c61 6365 2078 7920 7769 7468   replace xy with
000000e0: 200a 7468 6520 6e75 6d62 6572 206f 6620   .the number of
000000f0: 7468 6520 7374 6172 7469 6e67 2063 6c75  the starting clu
00000100: 7374 6572 206f 6620 7468 6973 2066 696c  ster of this fil
00000110: 6520 6173 2069 7420 6973 2070 7265 7365  e as it is prese
00000120: 6e74 2069 6e20 7468 6520 6469 7265 6374  nt in the direct
00000130: 6f72 7920 656e 7472 792e 200a 0000 0000  ory entry. .....
00000140: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000150: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000160: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000170: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000180: 0000 0000 0000 0000 0000 0046 4c41 477b  ...........FLAG{
00000190: 4953 5f54 4849 535f 5245 414c 7d00 0000  IS_THIS_REAL}...
000001a0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001b0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001c0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001d0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
:
```

The flag in the slack space is FLAG{IS_THIS_REAL}

I used Use fls to list files and identify potential slack space.

I got the inode as 5
dd if="image.dd" of=cluster.bin bs=512 skip=92 count=4
This extracts the clusters and svaes it to cluster.bin

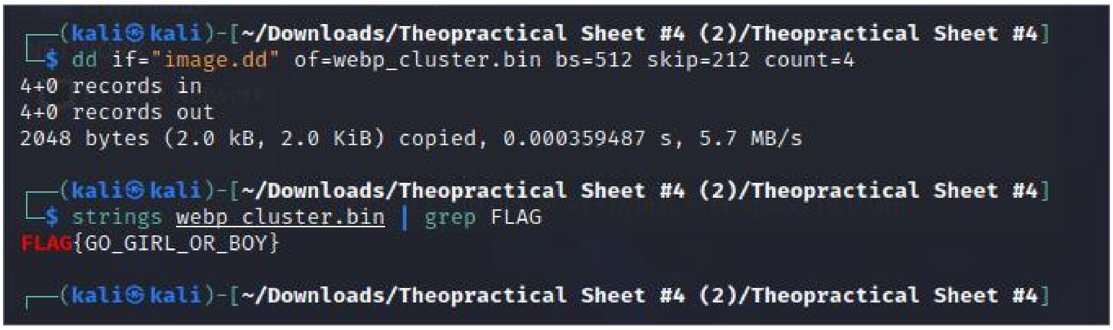
After dumping the cluster xxd cluster.bin > cluster_dump.txt
cat cluster_dump.txt | less

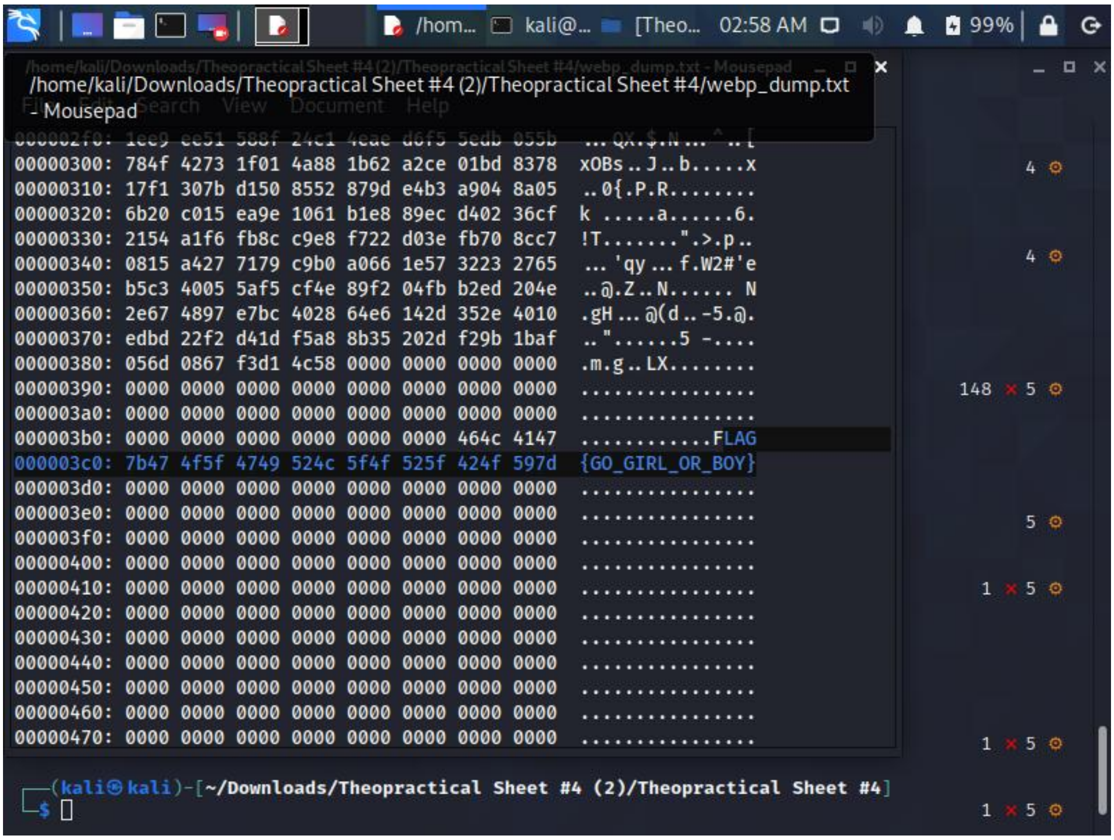
I got the FLAG{IS_THIS_REAL}

The flag FLAG{IS_THIS_REAL} was found within the same allocated cluster but after the actual file content was.This excess space is called as Drive Slack since it goes beyond the file's final sector and into the cluster's unused area.

Question 1.6 (1 point)
Find the flag in the slack space of the WEBP image (0.5 points). Is it part of the RAM Slack or the Drive Slack? (0.5 points)

Ans



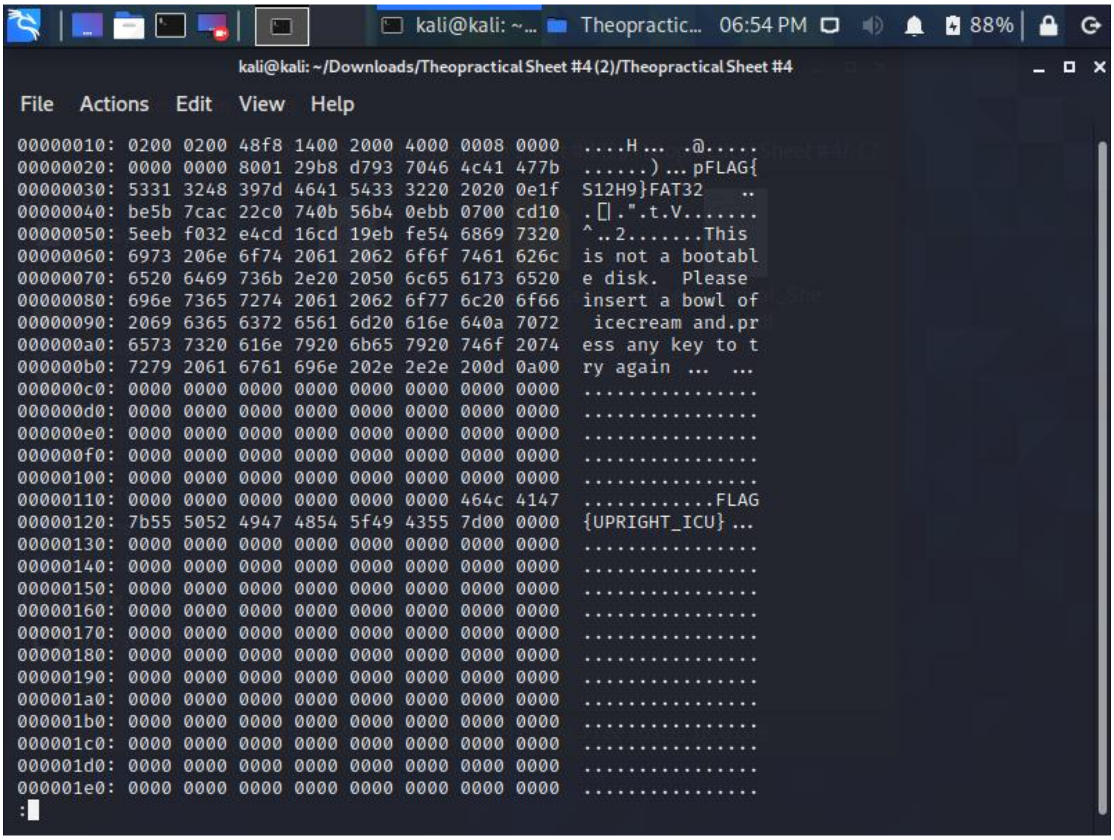


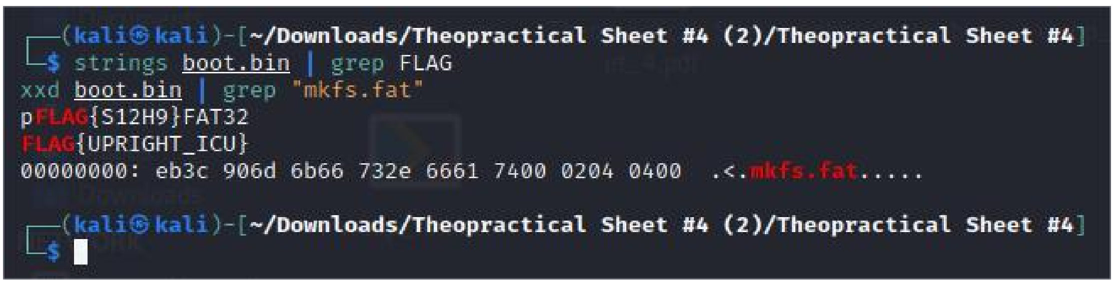
For the webp file I got the inode as 10 from the fls and istat output.
After extracting the clusters and saving it using dd command

I got 000003c0: 7b47 4f5f 4749 524c 5f4f 525f 424f 597d {GO_GIRL_OR_BOY}
I located FLAG{GO_GIRL_OR_BOY} in the cluster of the WEBP file at offset 0x03C0.It is located in RAM Slack (bytes 395–512, byte 960, sector 213).

Question 1.7 (1 point)
Find the flag in the non-essential part of the boot sector (0.5 points). On which operating system (Windows, macOS, Linux) was this partition created and formatted? You can assume that this value was not manipulated (0.5 points)

Ans



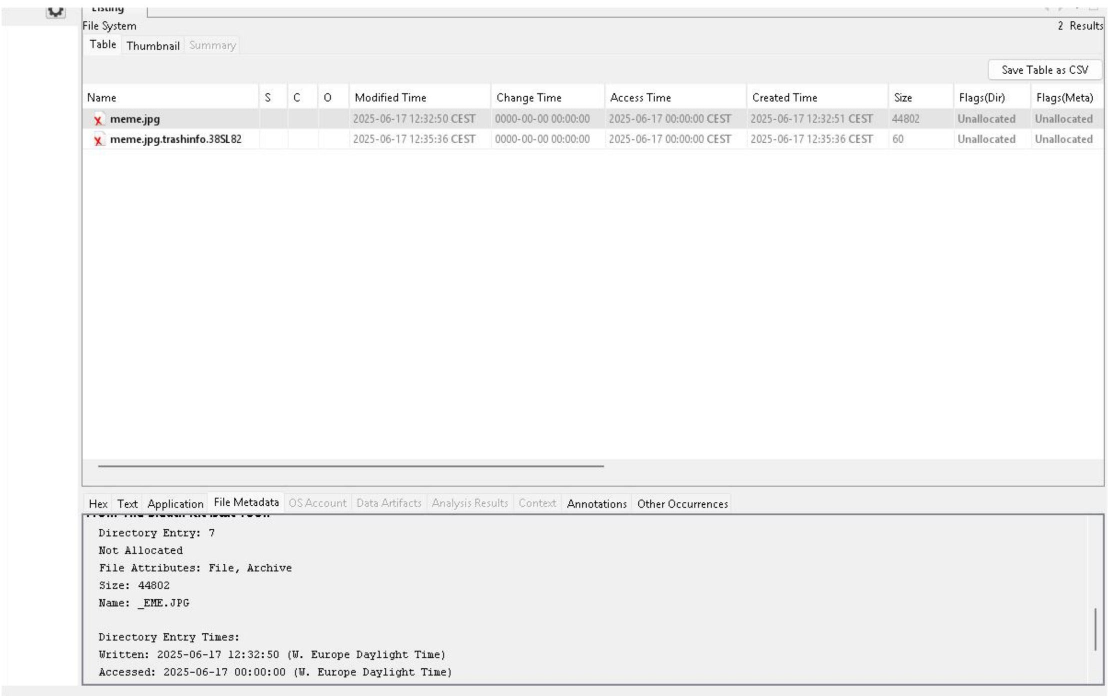


First I extracted the the boot sector using dd command after knowing the information from fsstat from the 1st question we know that OEM name is mkfs.fat
After using xxd to find the flag I got FLAG{UPRIGHT_ICU} at offset 0x11A.

mkfs.fat at offset 0x03 (OEM name) confirms Linux.

Question 1.8 (0.5 points)
How many bytes of “meme.jpg” were overwritten with other data? Give the result as a decimal number.

Ans:I used autopsy to get the file metadata
File size: 44,802 bytes
Allocated sectors: 80–83 (4 sectors)
Cluster size (from fsstat earlier from 1.1 question): 2048 bytes (4 sectors per cluster)



Since the file size is 44802 bytes and we know the eash cluster is 2048 bytes.
44802/2048=22 clusters.

Now we can get the total allocated space:
22*2048=45056 bytes

From The Sleuth Kit istat Tool:

Directory Entry: 7

Not Allocated

File Attributes: File, Archive

Size: 44802

Name: _EME.JPG

Directory Entry Times:

Written: 2025-06-17 12:32:50 (W. Europe Daylight Time)

Accessed: 2025-06-17 00:00:00 (W. Europe Daylight Time)

Created: 2025-06-17 12:32:51 (W. Europe Daylight Time)

Sectors:

Starting address: 80, length: 4

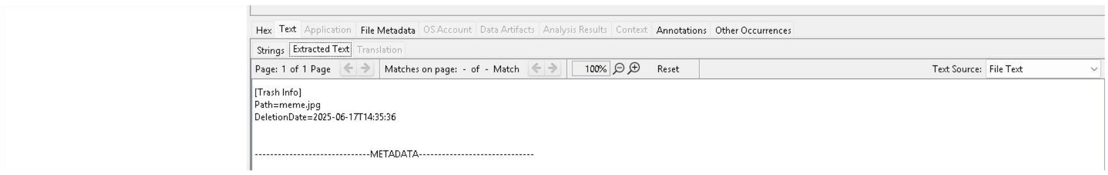
We know that autopsy recoved 4 sectors which is basically one cluster so this means the rest of the file data is overwriten.

45056-2048=43008 bytes

So 43008 bytes were overwritten

Question 1.9 (0.5 points)
When was the meme deleted? Give the timestamp in the format “yyyy-mm-ddThh:mm:ss”

Ans. Since we know user deletes a file in Linux the file isnt removed completely.
I found a transhinfo file in which the deletion is mentioned alond with the date.



[Trash Info]
Path=meme.jpg
DeletionDate=2025-06-17T14:35:36

Question Block 2 (4 points)
Question 2.1 (0.5 points)
Why do clusters in FAT always start at address 2?

- “Because clusters 0 and 1 are reserved for system use and metadata.”
- “Because the first two clusters are used to store the root directory.”
- “Because cluster 2 is the only one aligned with sector boundaries.”
- “Because clusters 0 and 1 are used to store the bootloader code”

Ans:Because clusters 0 and 1 are reserved for system use and metadata
Clusters starts at address 2

Question 2.2 (0.5 points)
What is the maximum number of characters in a short file name (SFN)? Give the answer as a decimal

Ans:Normally in 8.3 naming convention
8-character filename and 3-character suffix

8 + 3 = 11 characters

Question 2.3 (0.5 points)
A file has a valid FAT cluster chain but no corresponding directory entry. What does this suggest?

- “The file is a system file.”
- “It was never written.”
- “The directory entry was deleted or corrupted.”
- “The file is encrypted.”

Ans:The directory entry was deleted or corrupted.
Question 2.4 (0.5 points)
A file is deleted from a FAT file system. What happens to its directory entry?
Ans:The first byte is replaced with 0xE5.

If a directory entry (not a FAT entry) is unassigned, the first byte contains the value 0xE5 and also FAT marks delted entries with 0xE5.

Question 2.5 (0.5 points)
What hexadecimal value typically marks a bad cluster in the FAT?

- “0x00000000”
- “0x0FFFFFF7”
- “0xFFFFFFFF”
- “0x00000001”

Ans“0x0FFFFFF7”

0x0FFFFFF7 (FAT32): Cluster is damaged

Question 2.6 (0.5 points)
What can Alternate Data Streams (ADS) be used for?

- “Encrypting File Systems”
- “Extending file system capacity”

•
“Storing hidden content within a file”
•
“Repairing the $MFT”

Ans:“Storing hidden content within a file”

Question 2.7 (0.5 points)
What happens when a file is deleted in NTFS?
•
“Its $DATA is immediately wiped.”
•
“The in-use flag in the MFT is cleared.”
•
“Its MFT entry is zeroed out.”
•
“The $MFTMirr is updated”

Ans:“The in-use flag in the MFT is cleared.”

If an MFT entry is deleted, only the header is changed, all other data remains unchanged
• MFT entries are usually reused

Question 2.8 (0.5 points)
Which of the following timestamps is not visible in the Windows GUI?
•
“Creation Time”
•
“Modified Time”
•
“Accessed Time”
•
“MFT Modified Time”

Ans: We know that most important information in $STANDARD_INFORMATION = Timestamps:
MFT Modified Time not displayed in properties.

Culminative JSON:
{
 "name": "Thota Vijay Kumar",
 "matriculation": "7056597",
 "question_block_1": {
  "question_1_1": "FLAG{S12H9}",
  "question_1_2": "FAT16",
  "question_1_3": "44",
  "question_1_4": "FLAG{LETS_GET_STARTED_AT_CLUSTER_06}",
  "question_1_5": {
   "flag": "FLAG{IS_THIS_REAL}",
   "slack": "Drive Slack”
  },

"question_1_6": {
"flag": "FLAG{GO_GIRL_OR_BOY}",
"slack": "RAM Slack"
},
"question_1_7": {
"flag": "FLAG{UPRIGHT_ICU}",
"operating-system": "Linux"
},
"question_1_8": "43008",
"question_1_9": "2025-06-17T14:35:36"
},
"question_block_2": {
"question_2_1": ["Because clusters 0 and 1 are reserved for system use and metadata."],
"question_2_2": "11",
"question_2_3": ["The directory entry was deleted or corrupted."],
"question_2_4": ["The first byte is replaced with 0xE5."],
"question_2_5": ["0x0FFFFFF7"],
"question_2_6": ["Storing hidden content within a file"],
"question_2_7": ["The in-use flag in the MFT is cleared."],
"question_2_8": ["MFT Modified Time"]
}
}